$^{40}Ar/^{39}Ar$ phlogopite and U–Pb perovskite dating of lamprophyre dykes from the eastern Lake Superior region: evidence for a 1.14 Ga magmatic precursor to Midcontinent Rift volcanism

M. Queen, L.M. Heaman, J.A. Hanes, D.A. Archibald, and E. Farrar

Abstract: Thin lamprophyre dykes are abundant in the vicinity of the ca. 1.1 Ga Midcontinent Rift System and the late Archean – Early Proterozoic Kapuskasing Uplift in the Canadian Shield northeast of Lake Superior. However, the relationship between these dykes and spatially associated alkalic-carbonatite complexes has been unclear. To ascertain the temporal relationship between the two, we have determined, by $^{40}Ar/^{39}Ar$ and U–Pb dating, precise emplacement ages for six lamprophyre dykes from the region. The 1144^{+6}_{-4} Ma U–Pb date for perovskite from the western Marathon and Wawa areas is indistinguishable from the 1144 ± 7 Ma $^{40}Ar/^{39}Ar$ date for phlogopite from the eastern Kapuskasing Uplift, and it is concluded that these dykes are part of a regional 1.14 Ga swarm that extends over 300 km. These dykes were therefore emplaced some 35 Ma prior to the 1109 Ma commencement of Midcontinent Rift volcanism and coeval alkaline-complex activity, but essentially simultaneously with emplacement of the 1141 ± 2 Ma Abitibi diabase dyke swarm. It is suggested that these widespread 1144 Ma lamprophyre dykes and the Abitibi diabase dykes represent the first magmatic response of initiation of the Midcontinent rifting event, and were precursors to the main rift magmatism at 1109 Ma.

Résumé : Les dykes de lamprophyre sont abondants au voisinage du système du Rift Midcontinent, âgé approximativement de 1,1 Ga, et du soulèvement de Kapuskasing, d'âge Archéen tardif – Protérozoïque précoce, dans le Bouclier canadien, au nord-est du lac Supérieur. Cependant, les relations entre ces dykes et les complexes de carbonatites – roches alcalines spatiallement associés sont mal connues. Afin de mieux établir les relations chronologiques qui existent entre eux, nous avons déterminé, au moyen de datations $^{40}Ar/^{39}Ar$ et U–Pb, les âges précis de mise en place de six dykes de lamprophyre de cette région. La date de 1144^{+6}_{-4} Ma fournie par la méthode U–Pb sur le minéral pérovskite provenant des régions occidentales de Marathon et de Wawa est identique à la date de 1144 ± 7 Ma obtenue par les analyses $^{40}Ar/^{39}Ar$ sur phlogopite de la région orientale du soulèvement de Kapuskasing; par conséquent, nous tirons la conclusion que tous ces dykes appartiennent à un essaim régional vieux de 1,14 Ga, qui s'étend sur plus de 300 km. Ces dykes ont donc été mis en place quelque 35 Ma avant les débuts du volcanisme et de l'activité contemporaine qui a produit les complexes alcalins, datés de 1109 Ma et associés au Rift Midcontinent, cependant la mise en place des dykes est essentiellement contemporaine de celle des dykes de diabase dans l'essaim d'Abitibi vieux de 1141 ± 2 Ma. Il est proposé que les dykes de lamprophyre et les dykes de diabase d'Abitibi, largement répandus et âgés de 1144 Ma, représentent la première réponse magmatique du début de l'ouverture du Rift Midcontinent, et qu'ils ont servi de précurseurs à l'activité magmatique principale dans le rift, datée de 1109 Ma.
[Traduit par la rédaction]

Introduction

A number of ca. 1.1 and ca. 1.9 Ga alkalic-carbonatite complexes are associated with the Midcontinent Rift System and the Kapuskasing Uplift in the Canadian Shield north of Lake Superior (Fig. 1). In the same region, thin lamprophyre dykes are common, and abundant near 1.1 Ga alkalic-carbonatite complexes. However, previous dating attempts have yielded a wide range (1.7–0.3 Ga) of apparent emplacement ages for these dykes, and it is unclear whether they have a direct link with the alkalic-carbonatite complexes. To ascertain the temporal relationship between the two, we have determined, by $^{40}Ar/^{39}Ar$ phlogopite and U–Pb perovskite dating, precise emplacement ages for six lamprophyre dykes from this region.

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M. Queen, J.A. Hanes,[1] D.A. Archibald, and E. Farrar. Department of Geological Sciences, Queen's University, Kingston, ON K7L 3N6, Canada.
L.M. Heaman.[2] Jack Satterly Geochronology Laboratory, Department of Geology, Royal Ontario Museum, 100 Queen's Park, Toronto, ON M5S 2C6, Canada.

[1] Corresponding author (e-mail: hanesj@geol.queensu.ca).
[2] Present address: Department of Geology, University of Alberta, Edmonton, AB T6G 2E3, Canada.

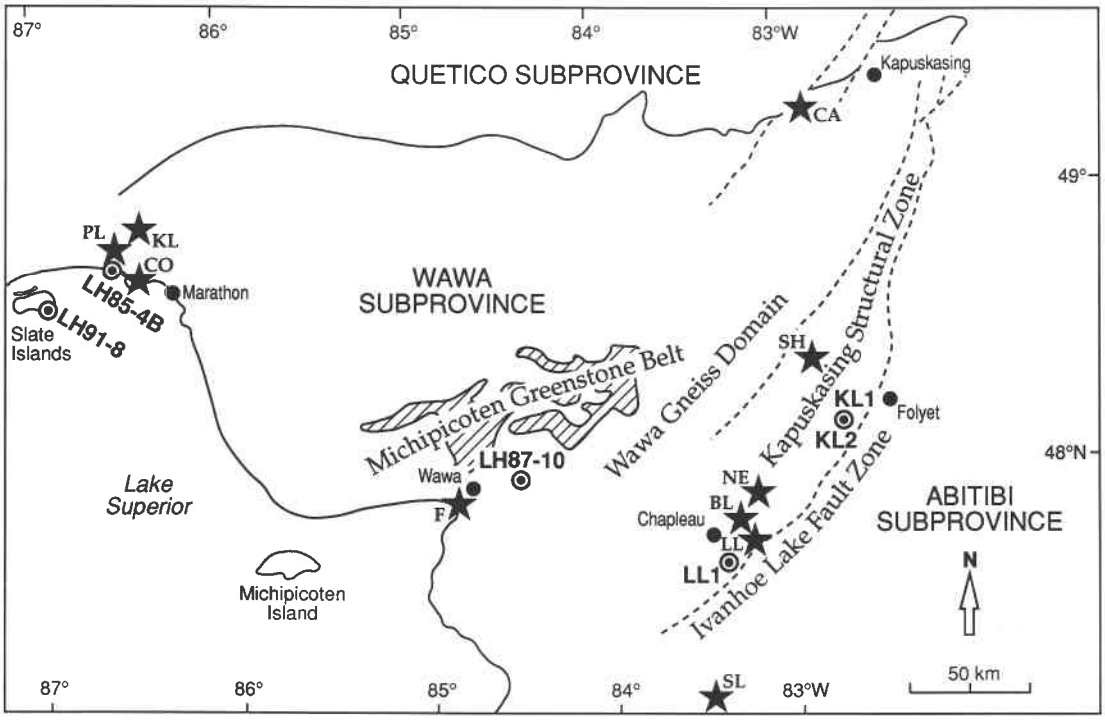
Geology and previous geochronological studies

Northeast of Lake Superior, the Archean Superior Province of the Canadian Shield is flanked by the 1.1–1.0 Ga Midcontinent Rift System to the southwest, the 1.8–1.7 Ga Southern Province to the south, and the 1.2–1.0 Ga Grenville

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Fig. 1. Sample locality map. ⊙, lamprophyre dykes; ★, alkalic-carbonatite complexes (CA, Cargill; SH, Shenango; NE, Nemegosenda; BL, Borden; LL, Lackner Lake; F, Firesand River; CO, Coldwell; KL, Killala Lake; PL, Prairie Lake). Broken lines are faults.



Province to the southeast. In this region of the Superior Province, the east–west-trending Wawa and Abitibi subprovinces are transected by the northeast-trending Kapuskasing Uplift, which is comprised of the granulite-grade Kapuskasing Structural Zone to the east and the amphibolite-grade Wawa Gneiss Domain to the west (Fig. 1). The high-grade rocks of the Kapuskasing Uplift were juxtaposed against the greenschist-grade Abitibi belt along the Ivanhoe Lake Fault Zone by ca. 1.9 Ga (Fig. 1) (Percival and Card 1985; Percival and West 1994). The Midcontinent Rift System to the southwest is an aborted Proterozoic continental rift that is exposed in the Lake Superior basin as the dominantly volcanic Keweenawan Supergroup and an associated intrusive suite that includes the Duluth Gabbro, the Logan Sills, and the Coldwell complex. U–Pb zircon ages bracket the rift volcanism to a 21 Ma period between 1108 and 1087 Ma (Davis and Sutcliffe 1985; Palmer and Davis 1987; Davis and Paces 1990).

Spatially associated with the Kapuskasing Uplift and Midcontinent Rift System are Proterozoic alkalic-carbonatite complexes of two ages. There are only a few ca. 1.9 Ga complexes, all in the Kapuskasing Uplift: the Cargill, Goldray, and Argor complexes (1906 ± 30, 1884 ± 2, and ca. 1950 Ma, respectively; Kwon 1986) in the north, and the Borden Township complex (1894 ± 29 Ma; Bell et al. 1987) in the south. Commoner are the ca. 1.1 Ga complexes, which occur in two areas (Fig. 1): (*i*) a north–south-trending belt on the northern flank of the Midcontinent Rift, associated with the Trans-Superior Tectonic Zone (Coldwell, Killala, Prairie Lake, and Chipman Lake); and (*ii*) a north-northeast-trending belt, associated with the Kapuskasing Uplift (Shenango, Nemegosenda, Lackner Lake, Firesand River, and Seabrook Lake). Precise U–Pb dates (Heaman and Machado 1992) provide accurate emplacement ages for three of the 1.1 Ga complexes: Coldwell (1108 ± 1 Ma), Lackner Lake (1105 Ma), and Nemegosenda (1107 Ma).

Lamprophyre dykes are a minor but ubiquitous component of the eastern Lake Superior region and can be traced for more than 300 km from west of Marathon to Folyet (Fig. 1), but their relationship to the alkalic complexes is unclear. In the Kapuskasing Uplift, the dykes are generally less than 1 m wide, with sharp contacts and chilled margins. They have vertical dips, but variable strikes, generally east to northeast. They cut all Archean rocks and diabase dykes of the Matachewan–Hearst (2.45 Ga; Heaman 1988) and Kapuskasing (2.05 Ga; Queen et al. 1989; Queen 1993) swarms, but they have not been observed to cut the ca. 1.1 Ga alkalic complexes. Thurston et al. (1977) suggested a Mesozoic age for these lamprophyres, based on a 131 ± 18 Ma K–Ar date (Wanless et al. 1973) for a dyke near Coral Rapids. A K–Ar biotite date of 1144 ± 33 Ma by Stevens et al. (1982) from a dyke near Folyet indicated that at least some of these dykes could be related to the ca. 1.1 Ga alkalic complexes, as was suggested by Percival (1981) based on their regional distribution.

Around Marathon (Fig. 1), in the western part of the study area, the diabasic and ultrabasic lamprophyric Marathon dykes (Platt and Mitchell 1979, 1982) are of two types: those that transect the 1108 Ma Coldwell complex, and those that cut the Archean Schreiber – White River greenstone belt but not the Coldwell (Platt et al. 1983). This paper deals with the second type. The lamprophyre dykes are thin (<1 m wide) and vertically dipping, with a predominant north–south

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strike (Platt and Mitchell 1982). Platt et al. (1983), based on a 1.65 ± 0.12 Ga Rb–Sr whole-rock isochron date, suggested that the dykes could be related to the ca. 1.7 Ga carbonatites in the Kapuskasing Uplift.

Lamprophyre dykes from the Slate Islands (Fig. 1) were interpreted to have a large range of ages (Sage 1991). A dyke on the north shore of Delaute Island is cut by a Keweenewan diabase dyke, showing that some lamprophyres are older than ca. 1.1 Ga. However, a K–Ar phlogopite date of 282 ± 11 Ma from a dyke (sample number 68 in Sage 1991) led to the conclusion that many of them are Pennsylvanian in age.

Sample description and analytical methods

We have dated six lamprophyre dykes sampled along an approximately 300 km transect, from the southeastern Kapuskasing Uplift near Folyet to the Slate Islands (Fig. 1).

Phlogopite from three dykes in the Kapuskasing Uplift were analysed by the incremental $^{40}Ar/^{39}Ar$ step-heating method. Two of these dykes (KL1 and KL2) are from an outcrop on Highway 101, approximately 30 km southwest of Folyet (Fig. 1). The dykes, 1 m and 60 cm wide, respectively, strike 048° with vertical dips, and cut granulite-grade tonalite gneisses (dyke KL1 also cuts a 2.05 Ga Kapuskasing dyke). The third dyke (LL1) cuts amphibolite-grade gneisses on Highway 129, approximately 20 km south of Chapleau (Fig. 1). The dyke is 30 cm wide and strikes 048°, with a vertical dip. Sample KL1 consists of handpicked phlogopite up to 5 cm in diameter. Samples KL2 and LL1 consist of smaller grains (<1 mm) obtained by standard mineral-separation techniques.

The three lamprophyre samples (Wawa, Marathon, and Slate Islands) used for the U–Pb perovskite study weighed 1.8, 0.5, and 5.9 kg, respectively. The Wawa sample (LH87-10), collected from a Highway 101 outcrop approximately 20 km northeast of Wawa, intrudes mafic gneisses and tonalites of the Wawa Gneiss Domain (Fig. 1). This approximately 10 cm wide composite dyke consists of subhedral to euhedral pale brown phlogopite, euhedral perovskite, Fe–Ti oxides, apatite, calcite clusters, serpentine clusters, and other unidentified fine-grained secondary minerals. A small, extensively altered, lamprophyre veinlet with a trachytic-like texture cuts the dyke, but phlogopite, perovskite, and Fe–Ti oxides can still be recognized. This veinlet was not removed prior to crushing.

The Marathon sample (LH85-4B) is from a site (C698) near McKellar Harbour (48°48′N, 86°44′W), approximately 40 km west of Marathon, studied in detail by Platt and Mitchell (1979). It consists of olivine, phlogopite, andraditic garnet, calcite, perovskite, apatite, Zr-rich titanian garnet, and spinels. At this site, the dyke is approximately 1 m wide and composite. It consists of a main porphyritic phase (LH85-4B) with irregular patches of finer grained lamprophyre that is transected by a narrow (approximately 2 cm), fine-grained, olivine-absent dykelet (LH85-4A). The porphyritic facies (LH85-4B) was the material used in this study. This dyke was not used in the Rb–Sr study of Platt et al. (1983), but microprobe analyses of perovskite were reported (Platt and Mitchell 1982).

The Slate Islands sample (LH91-8) is from a 30 cm wide, north–south-trending dyke on the southeast coast of Patterson Island. The dyke consists of olivine phenocrysts set in a matrix of garnet, phlogopite, magnetite, chlorite, serpentine, perovskite (up to 1 mm), calcite, apatite, and talc (Sage 1991). Sage (1991) analysed a sample (reference number 68) from this site for geochemistry, and obtained phlogopite for a K–Ar study.

Table 1. Results from the $^{40}Ar/^{39}Ar$ step-heating analyses for phlogopite from the lamprophyres.

(A) Integrated and plateau-segment dates.

Sample	Integrated date (Ma)	Steps on plateau	Plateau date (Ma)
KL1	1149±6	6–11	1148±6
		17–19	1140±7
KL2	1162±4	7, 8	1147±3
LL1	1157±20	6–11	1154±6

(B) Isotope correlation-plot results.

Sample	Steps	Calculated date (Ma)	$^{40}Ar/^{39}Ar$ ratio	MSWD
KL-1	3–23	1147±21	535±95	237.0
	6–11	1146±6	422±154	8.7
	17–19	1144±1	143±25	0.082
KL-2	2–4	1167±9	1867±517	0.51
	7, 8	1147±4	278±120	—
LL1	2–10	1146±4	541±86	1.95
	6–12	1154±24	296±24	2.3

$^{40}Ar/^{39}Ar$ method

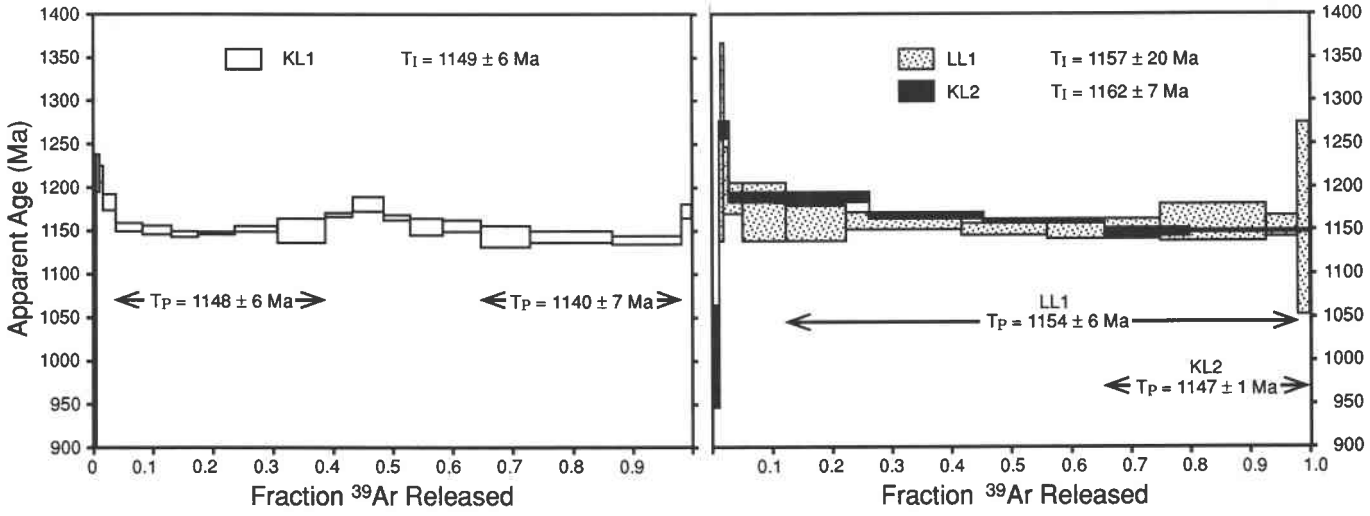
The methods used for the irradiation, fusion, and mass-spectrometric analysis of the samples and standards are the same as in Hanes et al. (1992). Two standards were used in this study: hb3gr hornblende (1072 Ma) and Obedjiwan biotite (962 Ma). System ^{36}Ar blanks varied from 0.06×10^{-10} cm^3 STP at 500°C to 0.09×10^{-10} cm^3 STP at 1150°C, and the argon had atmospheric composition. In the irradiation canister, there was a 1.5% variation along the length of the can, and the J values for the unknowns were interpolated based on their position in the can.

U–Pb perovskite method

Perovskite was isolated by standard crushing (jaw crusher, Bico disk mill), density (Wilfley table, methylene iodide), and magnetic (Frantz) separation techniques employing stringent cleaning procedures at all stages. All grains were examined with a binocular microscope at high magnification (×100) to select grains devoid of visible inclusions, fractures, and adhering material. The procedures for dissolving perovskite fractions in Savillex capsules and isolating U and Pb generally follow those outlined by Heaman (1989) and Heaman and Machado (1992). The analytical blanks for U and Pb average 1 and 10 pg, respectively. Sample solutions containing both U and Pb were loaded onto single Re filaments with Si gel and 0.5M H_3PO_4, and the isotopic compositions of U and Pb were determined using a VG354 mass spectrometer in single Faraday or Daly collector mode, depending on ion beam intensities. All isotopic ratios were corrected for mass discrimination of +0.10%/amu, and an

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Fig. 2. $^{40}Ar/^{39}Ar$ age spectra for phlogopite from lamprophyre dykes KL1, KL2, and LL1. The vertical widths of the boxes represent 2σ errors. T_I, integrated date; T_P, plateau date.



additonal correction of +0.3%/amu was applied to all ratios determined using the Daly photomultiplier detector. All 1σ errors reported in Table 2 were calculated by propagating all known sources of error, including a 1% uncertainty in the initial Pb isotopic composition, and are interpreted as minimum error estimates. Age errors are quoted at 2σ and were calculated with the two-error linear regression program of Davis (1982). The decay constants of ^{238}U (1.55125×10^{-10} year^{-1}) and ^{235}U (9.8485×10^{-10} year^{-1}) and the isotopic composition of uranium ($^{238}U/^{235}U = 137.88$) are the values determined by Jaffey et al. (1971) and recommended by Steiger and Jäger (1977).

Results

$^{40}Ar/^{39}Ar$ results

Results from the three $^{40}Ar/^{39}Ar$ step-heating analyses are presented in Table 1 and, as step-heating age spectra, in Fig. 2.[3] In the age spectra, the vertical heights of the bars for each fraction represent $\pm 2\sigma$ SE, suitable for comparing within-spectrum dates (i.e., error in $J = 0$). A conservative 0.5% error in J estimates can be used for between-spectra age comparisons of samples irradiated together; this corresponds to an additional ± 5 Ma (2σ) error in step dates.

The three phlogopite samples from the lamprophyre dykes in the Kapuskasing Uplift have similar integrated dates (Table 1), and their age spectra overlap to define a broad "plateau" date of ca. 1155 Ma (Fig. 2). However, the detailed structure of the age spectra permits a more accurate estimation of the phlogopite closure age.

The low dates in the first two steps of the KL1 spectrum, and in the first step of the KL2 and LL1 spectra, are attributed to minor argon loss from the phlogopites. All three spectra also contain some steps having dates above the ca. 1.15 Ga broad plateau. This is attributed to the presence of minor excess argon.

In detail, the age spectra of KL2 and LL1 are essentially identical (Fig. 2). In both, from the second step on, there is a monotonic decline in dates down to a plateau. For KL2, the last two steps define a plateau date of 1147 ± 1 Ma, which statistically overlaps the 1154 ± 6 Ma plateau date of LL1 for steps 6–11, inclusive (Fig. 2; Table 1). On isotope correlation plots of $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$, the data for each sample define two linear arrays. The higher temperature steps yield isochron dates that are close to the plateau dates, and $^{40}Ar/^{36}Ar$ intercept values that are atmospheric ratios (Table 1). For the lower temperature steps with elevated dates on the age spectra, the correlation dates are also close to the plateau dates, but the $^{40}Ar/^{36}Ar$ intercept shows high, non-atmospheric ratios (Table 1).

The more detailed (21-step) age spectrum of KL1 can be divided into two distinct U-shaped segments (Fig. 2), similar in character to the spectra of "two-faced" micas containing excess argon described by York and Lopez-Martinez (1986). The trough of the first segment (steps 6–11, inclusive) yields a plateau date of 1148 ± 6 Ma that is statistically indistinguishable from the 1140 ± 7 Ma plateau date for the trough of the second segment (steps 17–19, inclusive). Correlation dates for these same segments are 1146 ± 6 and 1144 ± 1 Ma, respectively (Table 1).

The plateau dates (Fig. 2), supported by the correlation dates (Table 1), provide the best estimate of the time of closure of the phlogopite to argon loss, and these ages statistically overlap. Since the ages are younger than the $^{40}Ar/^{39}Ar$ closure ages of both regional biotites (2.45–2.25 Ga; Hanes et al. 1994) and biotite (ca. 2.05 Ga; Queen et al. 1989) in the baked contact zone of the Kapuskasing dyke cut by KL1 lamprophyre, it is concluded that the closure ages of the phlogopite provide the time of dyke emplacement. The best estimate of the emplacement age is the 1144 ± 7 Ma average of the plateau segments from the KL1 spectrum. This value is chosen because the KL1 step-heating run was the most detailed, and also because the sample had been handpicked rather than having been obtained by crushing. Crushing could have led to difficulty in isolating radiogenic gas from excess argon for samples KL2 and LL1. This 1144 ± 7 Ma

[3] Complete isotope data may be purchased from the Depository of Unpublished Data, Document Delivery, CISTI, National Research Council Canada, Ottawa, ON K1A 0S2, Canada.

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Table 2. U–Pb perovskite results from Superior Province lamprophyre dykes.

Description[a]	Wt. (mg)	Pb (ppm)	U (ppm)	Th (ppm)	Pb_c^T (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	% Dis
Marathon (LH85-4B)													
1 M1.0A,d,br(137)	0.046	93	170	1244	211	435	0.18051±34	1.9370±61	0.07782±21	1070	1094	1142	6.9
2 M1.8A,c.dbr(200)	0.052	168	162	2719	229	446	0.18640±24	1.9963±57	0.07767±20	1102	1114	1139	3.5
3 NM1.8A,d,br	0.019	51	73	676	54	334	0.19591±49	2.1100±128	0.07811±42	1153	1152	1150	−0.3
4 M1.0A,b,bl(10)	0.009	188	228	2730	71	368	0.19341±42	2.0788±99	0.07795±32	1140	1142	1146	0.6
Wawa (LH87-10)													
5 NMO.8A,b,dbr(34)	0.068	41	83	457	204	349	0.19242±30	2.0528±74	0.07737±26	1134	1133	1131	−0.3
6 NMO.8A,d,dbr(140)	0.089	38	74	447	224	369	0.19096±30	2.0607±74	0.07827±26	1127	1136	1154	2.6
7 NMO.8A,c,bl(64)	0.055	35	67	415	120	383	0.19061±34	2.0435±70	0.07775±23	1125	1130	1141	1.5
8 NMO.8A,b,dbr(54)	0.090	45	86	538	229	414	0.18863±23	2.0282±63	0.07798±22	1114	1125	1146	3.1
Slate Islands (LH91-8)													
9 NM5,c,dbr	0.012	26	89	165	26	503	0.19166±40	2.0488±109	0.07753±36	1130	1132	1135	0.4
10 M1.7A,a,bl(1)	0.009	14	41	119	9	518	0.19212±72	2.0618±245	0.07784±86	1133	1136	1143	0.9
11 NM5,a,dbr(98)	0.128	25	70	220	180	611	0.19221±27	2.0573±49	0.07763±14	1133	1135	1137	0.4
12 NM5,c,gr	0.069	23	70	181	85	702	0.19243±33	2.0715±58	0.07807±18	1135	1139	1149	1.4

Notes: Th concentration estimated from the amount of ^{208}Pb and emplacement age. Atomic ratios corrected for spike, blank (10 pg Pb; 1 pg U), and initial common Pb following a model depleted mantle composition (Neymark 1990). Errors are quoted at 1σ. Pb_c^T, total common Pb; % Dis, percent discordance along a zero-age reference line.

[a]M1.0A refers to magnetic split on a Frantz isodynamic separator at 1.0 A and 10° side tilt. NM, nonmagnetic. Grain Size (mesh): a, +100; b, −100 + 200; c, −200 + 325. Colour: br, brown; dbr, dark brown; bl, black; gr, grey. Number in parentheses corresponds to the number of grains analysed.

Fig. 3. U–Pb concordia plot for perovskite from lamprophyre dyke LH85-4B.

age is indistinguishable from the average (1147 ± 6 Ma) of all the plateau segments from the three samples combined. The 1144 ± 7 Ma age from this study agrees with the 1144 ± 34 Ma K–Ar biotite date obtained by Stevens et al. (1982) on a lamprophyre from the Kapuskasing Uplift, but is more precise.

U–Pb perovskite results

The U–Pb results for four perovskite fractions from each of three lamprophyre samples are shown in Table 2 and on concordia diagrams in Figs. 3–5. The perovskite grains in all three samples are euhedral orange to dark brown cubes or octahedra. The U and Th concentrations in these perovskite fractions are quite variable, with the highest levels in the Marathon sample (73–228 and 676–2730 ppm, respectively), and the lowest in the Slate Islands sample (41–89 and 119–220 ppm, respectively).

The four fractions from the Marathon dyke (LH85-4B) are up to 7% discordant, but two fractions (LH85-4B, fractions 3 and 4, Table 2) are less than 1% discordant and overlap within error of the concordia curve (Fig. 3). Although this is the first reported case of perovskite discordance, its cause is unclear. Perovskite grains of differing magnetic susceptibility and size were selected, but there is no apparent correlation with the observed discordance pattern. A discordia line constructed to pass through all four analyses yields an upper intercept age of 1145^{+15}_{-10} Ma (64% probability of fit), with a lower intercept age of 90 Ma. This is interpreted as the best estimate for the emplacement age of the Marathon dyke, and indicates that it is not related to the ca. 1.9–1.7 Ga alkalic activity in the region, as was originally proposed by Platt et al. (1983) based on a 1650 Ma Rb–Sr date. Interestingly, Platt et al. (1983) also obtained a K–Ar whole-rock isochron date of 1120 ± 34 Ma that is in agreement with the U–Pb perovskite age from this study. We suggest that the Rb–Sr date is anomalously old.

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Fig. 4. U–Pb concordia plot for perovskite from lamprophyre dyke LH87-10.

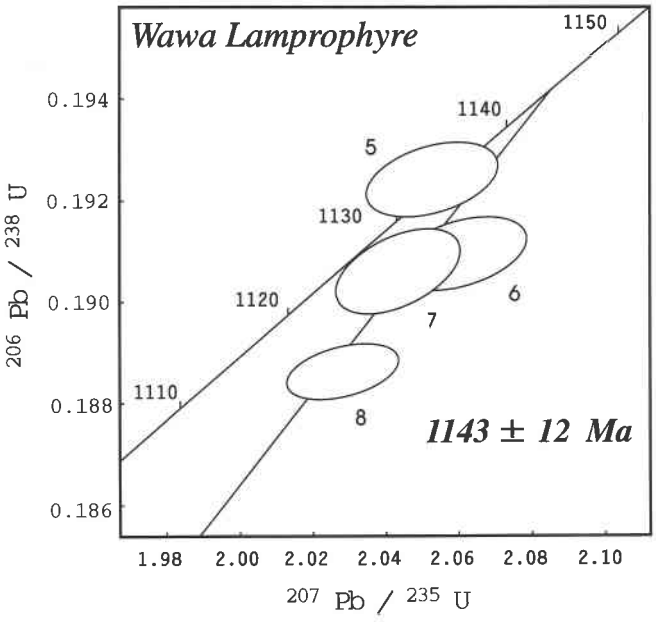
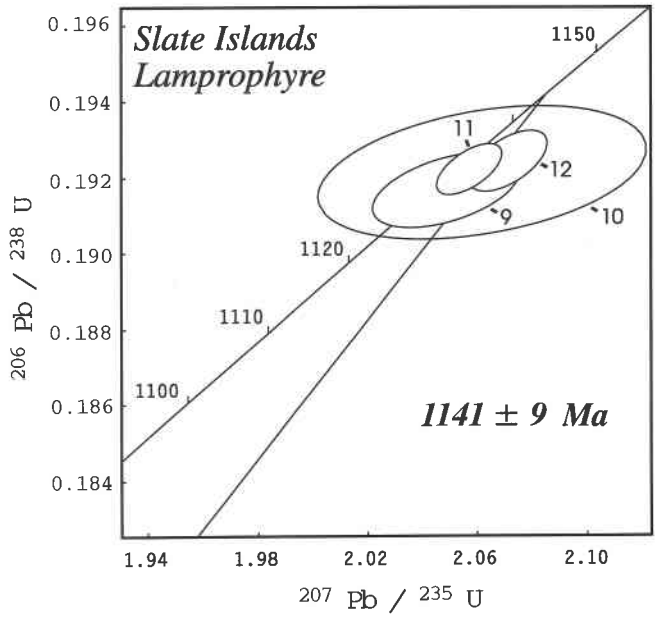


Fig. 5. U–Pb concordia plot for perovskite from lamprophyre dyke LH91-8.



The four perovskite analyses from the Wawa dyke (LH87-10) are up to 3% discordant, with a range of $^{207}Pb/^{206}Pb$ ages between 1131 and 1154 Ma (Table 2; Fig. 4). The Wawa analyses all overlap (Fig. 4) a 1144^{+6}_{-4} Ma composite reference discordia line as defined below. The average $^{207}Pb/^{206}Pb$ age of 1143 ± 12 Ma is interpreted as the best estimate for the emplacement age of the Wawa dyke.

All four analyses of perovskite from the Slate Islands lamprophyre (LH91-8), including a single-grain analysis (LH91-8, fraction 10), overlap within error of the concordia curve, with $^{207}Pb/^{206}Pb$ ages ranging from 1135 to 1149 (Table 2; Fig. 5). Also shown in Fig. 5 is the 1144^{+6}_{-4} Ma reference discordia line. The average $^{207}Pb/^{206}Pb$ perovskite date of 1141 ± 9 Ma is interpreted as the emplacement age of the Slate Islands lamprophyre. It is not clear why the K–Ar phlogopite date (282 ± 11 Ma) obtained previously at the same site (Sage 1991) is at variance with the U–Pb perovskite age. The Slate Islands and Marathon lamprophyres have mineralogical and geochemical similarities that are consistent with their similar U–Pb perovskite ages. For example, both Slate Island sample 68 (Sage 1991) and Marathon sample C698 (Platt and Mitchell 1979) consist essentially of olivine, phlogopite, garnet, calcite, perovskite, apatite, and spinel, ±serpentine and ±chlorite with a similar geochemistry, except that the Marathon sample has noticeably higher CaO, P_2O_5, Ba, and Sr contents. Furthermore, the white polycrystalline aggregates of garnet in the Slate Islands sample may have an origin similar to that of the andraditic garnet in the Marathon dyke that forms pseudomorphs after a lath-shaped mineral, possibly melilite.

Within uncertainties, the U–Pb perovskite ages obtained for all three lamprophyres are indistinguishable. A discordia line constructed using all perovskite analyses but fraction 5 (Table 2) gives an upper intercept age of 1144^{+6}_{-4} Ma (45% probability of fit; lower intercept age 64 Ma). This composite age is taken as the best estimate for the time of lamprophyre emplacement in the Wawa–Marathon region.

Discussion and conclusions

The 1144^{+6}_{-4} Ma U–Pb date for perovskite from the Marathon and Wawa lamprophyres is statistically indistinguishable from the 1144 ± 7 Ma $^{40}Ar/^{39}Ar$ date for phlogopite from the lamprophyres in the eastern Kapuskasing Uplift. These dates are interpreted as emplacement ages of the dykes, and it is thus concluded that there is a regional swarm of 1.14 Ga lamprophyres that extends over 300 km, from the Marathon region to Folyet (Fig. 1). The variation in dyke trend from northerly in the Marathon region to more northeasterly in the Kapuskasing Uplift creates a fan pattern centred on the Lake Superior basin, which suggests radiation of the dykes from a centre of rifting.

The precise age of these lamprophyre dykes is, however, some 35 Ma older than both the U–Pb 1109 Ma age for the commencement of Midcontinent Rift volcanism (Davis and Sutcliffe 1985), and the U–Pb ages for three of the ca. 1.1 Ga alkalic-carbonatite complexes in the region (Heaman and Machado 1992), namely Coldwell (1108 ± 1 Ma), Lackner Lake (1107 Ma), and Nemegosenda (1105 Ma). Heaman and Machado (1992) argued that the alkaline-complex activity and early tholeiitic magmatism were coeval and represented the commencement of rifting, but the lamprophyre dykes are not contemporaneous with this magmatic activity.

Interestingly, the age of 1144 Ma for the lamprophyre dykes from this study is statistically indistinguishable from the 1141 ± 2 Ma age of the Abitibi diabase dyke swarm (Krogh et al. 1987). The Abitibi dykes, weakly alkaline in character (Ernst and Bell 1992; Ernst et al. 1987), occur mainly in the Abitibi subprovince to the east of the Kapuskasing Uplift, where they trend east-northeasat. However, Abitibi dyke trends are more northerly in the Kapuskasing Uplift and in the region north of Lake Superior, appearing also to fan

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out from the Lake Superior basin (Osmani 1991). This swarm is dominated by the Great Abitibi Dyke, which is up to 250 m wide and traceable for approximately 600 km, and terminates in the west near the easternmost cluster of lamprophyres dated in this study. Thus, the similarity in age and local trend indicates a possible connection between the Abitibi dykes and the lamprophyres.

We thus suggest that the Abitibi diabase dyke swarm and the widespread 1144 Ma lamprophyre dykes in the Lake Superior region represent the first magmatic response of the initiation of the Midcontinent rifting event, and that they were thus a precursor to the main rift magmatism that commenced some 35 Ma later at 1109 Ma. The initiation of the rifting may have been related to collisional events in the Grenville Province to the southeast, as argued by Ranalli and Ernst (1986). The lamprophyre dykes, in a manner similar to the younger ca. 1109–1102 Ma alkalic-carbonatitic complexes, appear to be strongly concentrated in zones of previous crustal weakness (the Kapuskasing Uplift and the Trans-Superior Tectonic Zone) near the rift centre. Some lamprophyre dykes near the Ivanhoe Lake Fault Zone on the eastern margin of the Kapuskasing Uplift have been mildly deformed and sheared, indicating that there has been minor movement on this structure since 1144 Ma.

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